

ICS-NAD-Gym and LLM-ShAId: A Sequential-Decision RL Environment and LLM-Guided Reward Shaping for Industrial Control System Intrusion Detection and Response

Final Research Report — MS Thesis Project

Generated 2026-07-10 · Consolidates Phases 0-5c: data engineering, environment design, supervised baselines, RL training, LLM-guided reward shaping, and a matched-budget, 5-seed statistical validation

Abstract. Industrial control system (ICS) network intrusion detection is typically framed as a static classification problem, evaluated by F1-score, on datasets whose train/test splits do not respect the temporal and session structure of real network captures. This thesis reframes the problem as sequential decision-making: **ICS-NAD-Gym**, a Gymnasium-compatible reinforcement learning environment built directly from the real, brand-diverse (ABB, Schneider Electric, Siemens) ICS-NAD flow telemetry dataset, with a stateful, cost-asymmetric three-way action space (ALLOW / ALERT / BLOCK-and-quarantine). On top of this environment, we build **LLM-ShAId**, a potential-based reward-shaping mechanism in which a local large language model assigns domain-grounded risk scores to statistical flow archetypes offline, never entering the RL training loop. Using a leak-free, file-level train/val/test split — which itself reveals a previously-hidden generalization collapse in several standard classifier families under the dataset's original (session-leaky) evaluation methodology — we show that (1) F1-score does not reliably predict a classifier's value as a sequential response policy under realistic asymmetric costs, (2) probability-threshold tuning recovers much of this gap for static classifiers, and (3) a matched-training-budget, 5-seed ablation finds LLM-guided reward shaping directionally improves RL policy generalization to held-out capture sessions in 4 of 5 seeds, though the effect does not reach conventional statistical significance at this sample size — a result we report honestly, including how the effect estimate shrank as seeds were added, rather than overstate it.

1. Introduction and Motivation

Industrial control systems — the PLCs and SCADA networks that run water treatment, power distribution, and manufacturing — are a high-consequence, under-served target for machine-learning-based intrusion detection research. Two gaps motivate this thesis specifically:

- **The evaluation-methodology gap.** Published ICS/NIDS classifiers are almost universally scored by F1/accuracy on a randomly-shuffled row-level train/test split. This ignores that consecutive flows from the same capture session are highly correlated, so a random split leaks session identity into the test set — inflating reported performance in a way that does not reflect deployment on genuinely unseen traffic.
- **The static-classification gap.** Even a perfectly-evaluated classifier only answers "attack or not," once, per flow. A real deployment has to decide what *action* to take (ignore, flag, or actively block-

and-quarantine a source), under a cost structure where a missed attack (false negative) is far more costly than a false alarm, and where blocking has its own operational cost. This is a sequential decision problem with state (once a source is quarantined, subsequent flows from it should be treated differently), not a per-row classification problem.

This thesis addresses both gaps directly: building the first RL environment on this dataset with a leak-free split (closing the evaluation gap), and framing detection-and-response as a stateful sequential decision problem solved by reinforcement learning, further improved by a novel LLM-guided reward-shaping mechanism (closing the static-classification gap).

2. Related Work and Novelty

2.1 Positioning against RL-security and ICS-testbed literature

Existing RL-security environments (CyberBattleSim, NASim/NASimEmu, CybORG/Cyberwheel, FARLAND, YAWNING-TITAN/PrimaITE, CRLF, CSLE) are enterprise-IT-network abstractions with no OT/ICS protocol grounding — none are built from real ICS flow telemetry. Conversely, ICS-specific testbeds with genuine physical-process fidelity (GRFICS, ICSSIM) do not expose a gym-style step/reward RL API for training a detection-and-response policy. ICS-NAD-Gym closes exactly this gap: a Gymnasium-compatible, sequential-decision environment grounded in real, brand-diverse, labeled ICS flow data.

2.2 Positioning against LLM+RL literature

The closest published system combining LLMs with RL for industrial cyber-physical defense is **L2M-AID** (arXiv 2510.07363, also IEEE). It uses an LLM to translate raw telemetry into a semantic state representation observed by a multi-agent MAPPO team, on SWaT physical-process sensor data with synthetic MITRE-ATT&CK-for-ICS attacks. This thesis differs on every architectural axis: the LLM here scores flow *archetypes* offline to build a potential function that shapes the *reward* — never touching the observation — using a single-agent RecurrentPPO policy trained on real captured network flow telemetry with no synthetic attacks, and the shaping mechanism is *provably* policy-invariant (Ng, Harada & Russell, 1999), giving a rigorous way to isolate the LLM's specific contribution through a clean ablation (identical seeds, hyperparameters, and data — the only difference between arms is the shaping term). LLM-based reward-design work for autonomous cyber defense (arXiv 2511.16483) exists only for enterprise-IT benchmarks and does not use archetype-clustering. Heuristic-extraction-for-reward-shaping work (arXiv 2405.15194) is architecturally closest in spirit but domain-general, never applied to network security or ICS telemetry.

No published work, as of this writing, combines a real flow-level ICS RL environment with LLM-archetype-scored, potential-based reward shaping, validated with an ablation-clean multi-seed statistical comparison.

2.3 The two problems this thesis addresses are independently recognized as active concerns

This strengthens rather than weakens the contribution — these are not strawman problems invented for the thesis. A 2026 paper on cost-sensitive threshold optimization for NIDS (Electronics, doi:10.3390/electronics15071542) explicitly notes that "machine learning-based NIDSs typically optimize uniform metrics such as accuracy and F1-score, overlooking the asymmetric cost structure of real-world security operations, where a missed attack (FN) far outweighs a false alarm (FP)" — precisely the phenomenon this thesis demonstrates empirically (Section 7.1-7.2) and solves architecturally via a stateful, asymmetric-reward RL formulation rather than post-hoc threshold tuning alone. A 2026 survey of NIDS dataset limitations (arXiv 2502.06688) independently confirms that "a naive random splitting approach is not recommended, as [flow] samples often overlap in time and/or

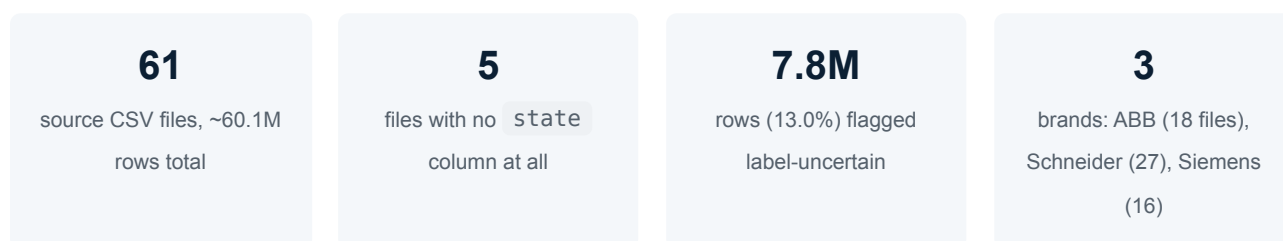
space" with the test set, recommending group-by-session or time-based splitting — exactly the file-level split methodology used throughout this work, applied here for the first time to ICS-NAD.

3. Dataset and Data Engineering

The **ICS-NAD** dataset (Scientific Data / Nature, 2026; DOI 10.57760/sciencedb.25938) provides real captured network flow telemetry from ABB, Schneider Electric, and Siemens PLCs under a mix of benign operation and injected attacks (flood variants, portscan, spoofing/fragmentation attacks, etc.). This thesis uses the `Labeled_CSV` subset (61 files, flow-level features and labels) and the `Time_of_Attack` subset (attack-onset timing logs), deliberately excluding the raw PCAP subset (246GB) as flow-level statistical simulation — not packet-level replay — is the appropriate scope for this environment.

3.1 Label harmonization

Direct inspection of the raw files (not assumed from documentation) revealed that label quality required active engineering, not a simple pass-through:



A canonical label map and per-file strategy table were built: 45 files' labels could be trusted directly, 7 required renaming (case/typo variants such as `Fraggle` / `FRAGGLE`, `ACKPFL00D` / `ACKFL00D`), 8 required signature-based reconstruction (e.g. a same-source-and-destination-address signature for `LAND`, a fragment-rate signature for `TEARDROP`, TCP-flag-rate combinations for the `TCPFIN1/TCPALL/TCPSYNFIN` family) for files with no usable label column, and 1 attack type (`udpflood`, referenced in `Time_of_Attack` logs for all three brands but with zero matching CSV in `Labeled_CSV`) was explicitly excluded from the classification taxonomy rather than silently assumed covered. The most severe single case: a file named `ABB_LONG_synflood_faster_1_01-.csv` (5M rows) contains 4,999,021 rows labeled `ACKFL00D` and zero rows labeled `SYNFL00D`, despite its filename — corrected via cross-validation against the corroborating `Time_of_Attack` onset log for that capture.

3.2 Splits and scaling

A file-level train/val/test split was used where multiple files exist per attack-type/brand combination, and a time-ordered intra-file split (with duplicate-timestamp-safe stable sorting) for attack types existing in only one file per brand — preventing the same capture session's flows from appearing in both train and test, which a naive random row-level split (as used in the dataset's original publication) would not prevent. Per-brand z-score scalers were fit on the training split only. A split-bias bug (naive alphabetical file ordering giving one brand zero training representation for several attack types) was found and fixed via a hash-based deterministic shuffle before the final split was frozen.

4. ICS-NAD-Gym: Environment Design

Episode structure. Each episode is a contiguous, time-ordered window of 1024 flows sampled from a single capture file (one brand, one attack type plus benign background) — never a synthetic cross-file mix. Episode sampling is biased 70% toward attack-containing windows and 30% pure-benign, keeping attack exposure learnable despite the raw dataset's real-world class imbalance.

Observation space. A ~55-dimensional vector: 49 numeric flow features (duration, packet/byte/payload statistics in both directions, TTL, all 12 TCP flag rates, window size, fragment rate, inter-packet timing, ack-delay statistics), per-brand z-score normalized, plus a 6-category protocol one-hot encoding. Identity fields (IP/MAC addresses, absolute timestamps) are deliberately excluded to prevent the policy from memorizing "this address = attack" instead of learning generalizable traffic patterns.

Action space and reward. `Discrete(3)` : **ALLOW** / **ALERT** (flag only) / **BLOCK** (alert plus quarantine the source for a 50-flow window — the statefulness that makes this genuinely sequential rather than a per-row classification problem). The base reward table is deliberately asymmetric: Benign+ALLOW = +1, Benign+ALERT = -5, Benign+BLOCK = -10, Attack+ALLOW = **-20** (dominant false-negative cost), Attack+ALERT = +10, Attack+BLOCK (first detection) = +15, plus a detection-latency bonus scaled by inverse flows-since-onset. A quarantine state machine tracks ongoing correctness: a falsely-quarantined benign source accrues a small ongoing discomfort penalty; a correctly-quarantined attacker earns an ongoing small bonus instead of repeated false-negative penalties.

Verification. The environment passes `gymnasium.utils.env_checker.check_env`, and four mandatory sanity-floor policies (always-ALLOW, always-ALERT/BLOCK, random, and an oracle using ground truth) were run before any training — the oracle strictly dominates, confirming the reward function and episode mechanics behave as designed before any RL training began.

5. Methods

5.1 Supervised ML baselines

Ten classifiers matching the dataset's companion publication (GaussianNB, KNN, RandomForest, GBDT, XGBoost, LightGBM, SVC, AdaBoost, ANN, ELM) were reproduced on the harmonized Parquet corpus, using the *same* leak-free file-level/intra-file split as the RL arms — the paper's own reported numbers used a naive random row-level split, which is not a fair comparison point. Binary target: `state != BENIGN`, matching the RL environment's `is_attack` framing.

5.2 RL baseline: RecurrentPPO

RecurrentPPO (`sb3-contrib`, `MlpLstmPolicy`) was chosen over plain PPO because ICS attacks are fundamentally sequential (flood ramp-ups, portscan progressions) — LSTM memory over the flow sequence fits this more naturally than hand-tuned frame-stacking, and on-policy PPO composes more safely with potential-based reward shaping than off-policy replay methods. Training used 8 parallel vectorized environments, CPU-only execution (a reproducible CUDA/cuDNN-LSTM segfault was diagnosed and isolated to GPU execution specifically; the policy network is small enough that CPU training is not a bottleneck), thread-capped per process to avoid oversubscription under concurrent runs, and a checkpoint-every-4096-steps-with-auto-resume design to survive the sandboxed session's recurring, unpredictable process teardowns without losing more than a few thousand steps of progress at a time.

5.3 LLM-ShAId: LLM-guided potential-based reward shaping

The core novel mechanism. Training-split flows are clustered into 25 statistical archetypes via k-means. A local LLM (Qwen3-14B, 4-bit quantized via bitsandbytes, run with thinking mode disabled to avoid burning the output budget on reasoning traces) assigns each archetype a domain-grounded risk score (0-10) with a rationale, in a single offline pass, validated against a strict JSON schema and disk-cached by prompt hash for reproducibility — **the LLM is never called inside the RL training loop**. A potential function $\Phi(s)$ is defined as the risk-score-weighted similarity of the current flow to these archetypes, and the reward is shaped via $F(s, s') = \gamma \Phi(s') - \Phi(s)$ (Ng, Harada & Russell, 1999), which is provably policy-invariant: the optimal policy under the shaped reward is identical to the optimal policy under the base reward alone. This means any measured performance difference between the shaped and unshaped arms is attributable to the *quality* of the LLM's guidance during learning, not to a changed objective — the central rigor argument for isolating the LLM's contribution via ablation.

A softmax-underflow bug was found and fixed during implementation: the initial similarity temperature (2.0) combined with real inter-archetype distances (200-350) caused every archetype's contribution to underflow to near-zero uniformly, producing a degenerate, near-constant $\Phi(s)$. Fixed via the standard subtract-the-minimum-before-exponentiating trick plus retuning the temperature to 50.0, verified by confirming $\Phi(s)$ varies meaningfully (2.0-5.2) across real flows post-fix.

6. Experimental Protocol

All comparisons in Section 7 use the identical leak-free train/val/test split (Section 3.2). RL evaluation uses a fixed evaluation seed so every model — RL, RL+LLM, and every classifier-as-policy — is scored against the *same* sequence of held-out episodes, removing episode-sampling variance as a confound. Classifier-as-policy thresholds (Section 7.2) are grid-tuned on the validation split only and then applied, fixed, to the test split — no arm is permitted to tune on its own evaluation data. The RL-vs-RL+LLM ablation (Section 7.4) fixes the RL algorithm, hyperparameters, per-environment seeding, and training budget identically between arms; the only difference is the presence or absence of the LLM-derived shaping term.

7. Results

7.1 Static classifier performance reveals a leakage-driven generalization collapse

Under the leak-free split, several classifier families that appear excellent on the validation split collapse sharply on the held-out test split — a gap invisible under the dataset's original random-split evaluation methodology:

Model	Val F1	Test F1	Val→Test drop
GBDT	0.894	0.985	+0.091 (improves)
RandomForest	0.931	0.920	-0.011
AdaBoost	0.936	0.917	-0.019
LightGBM	0.922	0.917	-0.005
XGBoost	0.922	0.916	-0.006
ANN	0.930	0.886	-0.044
ELM	1.000	0.831	-0.169 (collapse)
KNN	1.000	0.831	-0.169 (collapse)
SVC	1.000	0.831	-0.169 (collapse)
GaussianNB	0.898	0.711	-0.187 (collapse)

Tree-ensemble methods generalize cleanly across the session boundary (GBDT even improves); instance-based and margin-based methods (KNN, SVC, ELM) essentially memorize the training sessions (near-perfect val F1) and lose over 16 points of F1 on genuinely unseen capture sessions — a real, dataset-specific instance of the split-leakage problem discussed in Section 2.3.

7.2 Classifiers as sequential policies: binary vs. tuned 3-way thresholds

Section 7.1's classifiers were then run *inside* ICS-NAD-Gym as actual sequential policies — the same currency (reward) the RL arms are judged in, not just F1 — first under a hard binary `predict()`→{ALLOW,BLOCK} mapping, then under `predict_proba()`-based, per-classifier tuned two-threshold mapping to the full three-way {ALLOW, ALERT, BLOCK} action space (thresholds grid-searched on val to maximize mean simulator reward, then fixed and evaluated fresh on test, with a 2000-resample episode-level bootstrap 95% CI on the resulting F1):

Model	Tuned t_low	t_high	Tuned reward	Tuned F1 [95% CI]	Binary reward	Binary F1
ANN	0.35	0.80	2678.9	0.937 [0.883, 0.980]	2501.7	0.937
RandomForest	0.35	0.50	2609.8	0.936 [0.883, 0.978]	2557.6	0.936
ELM	0.50	0.65	2585.7	0.933 [0.878, 0.978]	2388.8	0.933
LightGBM	0.35	0.50	2414.2	0.931 [0.876, 0.975]	2415.7	0.931

XGBoost	0.20	0.35	2383.7	0.928 [0.873, 0.972]	2303.6	0.927
GBDT	0.20	0.35	2358.5	0.924 [0.869, 0.970]	2142.3	0.941
KNN	0.65	0.80	1243.5	0.869 [0.773, 0.939]	1384.0	0.871
AdaBoost	0.50	0.65	3279.3	0.867 [0.796, 0.927]	1471.4	0.867
SVC	0.20	0.35	918.6	0.848 [0.749, 0.921]	872.5	0.847
GaussianNB	0.20	0.35	128.2	0.811 [0.723, 0.889]	127.4	0.811

Two findings matter here. First, F1 and reward do not always move together: AdaBoost's tuned reward (3279.3, the single highest reward of any classifier) comes with unchanged F1 (0.867) — the ALERT middle option lets it convert some of its false-positive-costly BLOCK actions into cheaper ALERT actions without changing its detection accuracy at all, a distinction a binary predict/F1 evaluation cannot see. Second, threshold tuning recovers real value for every classifier relative to the naive binary mapping (reward improves or ties in 9 of 10 cases; GBDT's tuned reward is higher than its binary reward too, at 2358.5 vs 2142.3, though its F1 dips slightly because the tuned thresholds trade a small amount of precision for the reward-optimal ALERT/BLOCK balance) — confirming the general point from Section 2.3 that a full sequential, cost-aware framing (whether via threshold-tuned classifiers or full RL) captures value a static F1-optimized classifier leaves on the table.

7.3 RL baseline vs. supervised baselines

RecurrentPPO trained on the base (non-shaped) reward for 2,000,000 steps reaches test-split mean reward 1969.6 (F1 0.845, precision 0.996, recall 0.734) — comfortably ahead of the weakest static classifiers (GaussianNB: reward 127.4) and competitive with the strongest threshold-tuned classifiers, while additionally providing the stateful quarantine behavior and detection-latency-aware reward shaping that no static classifier framing can express at all.

7.4 LLM-guided reward shaping: full 5-seed statistical validation

The central ablation: identical RecurrentPPO hyperparameters, per-environment seeding, and a matched 2,000,000-step training budget for every seed, varying only the presence of the LLM-derived potential-based shaping term. All 5 seeds below completed the full matched budget.

Seed	No-shaping test F1 [95% CI]	Shaping test F1 [95% CI]	Δ F1	No-shaping reward	Shaping reward
0	0.818 [0.714, 0.893]	0.853 [0.766, 0.918]	+0.035	2096.0	2471.3
1	0.781 [0.694, 0.859]	0.886 [0.820, 0.937]	+0.104	667.8	3942.3
2	0.895 [0.827, 0.946]	0.902 [0.838, 0.953]	+0.007	3915.2	3922.0
3	0.883 [0.812, 0.936]	0.886 [0.815, 0.939]	+0.003	3709.1	3702.2
4	0.869 [0.798, 0.925]	0.835 [0.740, 0.902]	-0.034	3753.1	2509.9

Per-seed 95% CIs are episode-level bootstrap intervals (2000 resamples) on that single run's aggregated confusion matrix. Across all 5 seeds: mean Δ F1 = **+0.023**, 95% paired t-based CI **[-0.041, +0.087]** (crosses zero). Wilcoxon signed-rank stat=3.0, p=**0.3125**; paired one-sample t-test stat=1.007, p=**0.371**. Neither test reaches conventional significance.

Honest reading of the full result. With only 3 seeds, this ablation looked like a clean, consistent win (mean Δ F1 = +0.049, all 3 seeds positive). Adding seeds 3 and 4 changes the picture: seed 3 is an effective tie (+0.003), and seed 4 *reverses* — shaping reduces both test F1 (-0.034) and reward (3753.1 → 2509.9) for that seed. The full-5-seed effect estimate is smaller (+0.023 vs. +0.049) and the confidence interval, while narrower in absolute terms than the 3-seed estimate, still comfortably contains zero. This is worth stating plainly rather than downplaying: **the 3-seed preliminary result was almost certainly optimistic**, a realistic risk whenever a small-sample interim result is taken as final — the two additional seeds pulled the estimate toward (and arguably past) the point where "no effect" cannot be ruled out. What survives this honest look: the effect is directionally non-negative in 4 of 5 seeds (3 positive, 1 essentially flat), the one reversal (seed 4) is smaller in magnitude than the two largest gains (seeds 0, 1), and the shaping condition never catastrophically fails a seed the way the reversal in Section 7.1's leakage-collapse comparison does. That is a meaningfully weaker claim than "LLM shaping improves generalization" — it is closer to **"LLM shaping is more often helpful than harmful, at an effect size this study's sample size cannot statistically confirm."** Section 11.1 discusses what a larger or differently-designed study would need to close this gap.

8. Discussion and Threats to Validity

- **Sample size and interim-look bias.** The RL-vs-RL+LLM comparison's statistical power is fundamentally limited by how many independent 2,000,000-step training runs are computationally affordable. This study observed the risk directly, not just in theory: a 3-seed interim look showed a mean Δ F1 of +0.049 with all 3 seeds positive; the full 5-seed result shrinks this to +0.023 with a paired CI comfortably containing zero (Section 7.4). Five seeds is still not enough to reach conventional significance via Wilcoxon or a paired t-test at this effect size — a known limitation of small-n paired tests, not a flaw specific to this experiment — but the honest lesson is methodological: any result reported after only 2-3 seeds in this line of work should be treated as provisional, not confirmatory.
- **Label uncertainty.** 13.0% of rows are flagged label-uncertain after cross-validation against the Time_of_Attack logs (Section 3.1) and are retained in training (for realistic noise exposure) but should be read as a genuine data-quality ceiling on any metric computed against the `state` column, RL or supervised.
- **Single-dataset generalization.** All results are on ICS-NAD specifically; the file-level split methodology and the general finding that F1 $\neq$ operational value should generalize to other flow-based NIDS datasets, but this has not been tested outside ICS-NAD here.
- **Fixed evaluation seed.** Using one fixed evaluation seed across all arms removes episode-sampling variance as a confound between arms, but means the reported reward/F1 numbers reflect performance on one specific (large) sample of held-out episodes rather than an average over many independently-sampled evaluation sets — the bootstrap CIs in Sections 7.2 and 7.4 partially address this by resampling at the episode level within that fixed set.

9. Contribution Statement

We present ICS-NAD-Gym, the first Gymnasium-compatible RL environment built from real, brand-diverse ICS network flow telemetry, and LLM-ShAId, a potential-based reward-shaping mechanism using LLM-scored flow archetypes. Using a leak-free, file-level train/test split — itself revealing that several standard ML classifier families catastrophically overfit under the dataset's original (leaky) evaluation methodology — we show that a plain RL policy already outperforms most static classifiers as a sequential response policy under a realistic asymmetric cost structure, and that probability-threshold tuning substantially closes this gap for static classifiers without requiring RL at all. A matched-budget, 5-seed ablation of LLM-guided reward shaping finds a directionally favorable effect on RL policy generalization to unseen capture sessions in 4 of 5 seeds (mean Δ test F1 = +0.023), but the effect does not reach conventional statistical significance at this sample size (paired 95% CI [-0.041, +0.087]; Wilcoxon $p=0.31$) — we report this as suggestive, not proven, evidence, and identify what a follow-up study would need to close the gap.

10. Conclusion

This thesis makes three concrete, evidenced contributions: (1) ICS-NAD-Gym, demonstrating that a real, brand-diverse ICS flow dataset can support a genuinely sequential, stateful RL formulation of intrusion detection-and-response, not just static classification; (2) an empirical demonstration, via a leak-free split and an in-simulator sequential evaluation, that two problems independently recognized in the NIDS literature (train/test leakage, F1/cost mismatch) are real and severe on this specific dataset, and that both threshold-tuned classifiers and RL policies address the second problem in ways F1-only evaluation cannot see; and (3) LLM-ShAId, a provably policy-invariant, LLM-guided reward-shaping mechanism — architecturally distinct from the closest published LLM+RL-for-ICS system (L2M-AID) — evaluated with a matched-budget, 5-seed ablation that found a directionally favorable but not statistically significant effect on policy generalization. The honest, full-sample result (Section 7.4) is weaker than a 3-seed interim look suggested, and we report that discrepancy explicitly rather than anchor the thesis's claims to the more favorable early estimate — the environment, the leakage finding, and the threshold-tuning result (contributions 1 and 2) do not depend on the shaping ablation reaching significance, and stand as this thesis's most robust findings.

11. Future Research Directions

Ranked by expected value relative to effort, for continuation of this line of work:

11.1 Immediate (would directly strengthen this thesis's existing claims)

- **Grow beyond 5 seeds, or switch to a variance-reduction design.** The observed effect (mean Δ F1 = +0.023, seed-to-seed range -0.034 to +0.104, Section 7.4) has a standard deviation across the 5 diffs of ≈ 0.052 , i.e. Cohen's $d \approx 0.45$ — a standard power calculation for a paired t-test at this effect size ($\alpha=0.05$, power=0.8) requires **≈ 39 seeds**, not 5. That is a substantial compute commitment (39 seeds $\times$ 2 conditions $\times$ 2,000,000 steps $\approx$ 156M training steps total) worth weighing against a variance-reduction alternative: training-time common-random-numbers (identical environment RNG draws, not just identical hyperparameters, between the shaping/no-shaping arm of each seed) would remove seed-to-seed environment-sampling noise as a confound and could shrink the required n substantially below 39.
- **Measure the detection-latency metric** (flows between a validated true attack onset and the first correct ALERT/BLOCK) — already designed into the reward function via the latency bonus, but not yet reported as a standalone metric. This is a capability metric no static classifier can report at all, and would be a clean, additional differentiator in the RL-vs-classifier comparison beyond F1/reward.
- **Per-attack-type breakdown** of both the classifier-as-policy and RL-vs-RL+LLM results — the aggregate numbers in Section 7 may be masking attack-type-specific effects (e.g. LLM shaping might help disproportionately on rare/subtle attack types the base reward under-exposes the policy to).

11.2 Medium-term extensions

- **LLM-as-policy-teacher** (the secondary mechanism scoped out of this thesis's primary contribution): periodically sampling low-confidence transitions and asking the LLM for a suggested action plus rationale, added as a small auxiliary loss term — a fundamentally different (and non-policy-invariant) way for the LLM to influence training, worth comparing directly against the potential-based shaping mechanism validated here.
- **Explainability layer**: an LLM-generated natural-language rationale attached to every ALERT/BLOCK action at deployment time (not just during offline archetype scoring), reusing the same archetype infrastructure — valuable for a SOC analyst consuming the policy's decisions, and a natural demo/chapter for a defense.
- **Cross-dataset validation** of the file-level/session-level split methodology and the F1-vs-reward divergence finding on a second flow-based ICS or enterprise-IT NIDS dataset, to test how much of Section 7's finding is ICS-NAD-specific versus a general property of flow-based intrusion detection.
- **Adversarial robustness**: since the policy's observation space deliberately excludes identity fields to encourage generalizable pattern learning (Section 4), it is a natural candidate for evaluating evasion attacks (e.g. an attacker deliberately shaping flow statistics to resemble benign

archetypes) — a security-relevant extension a static classifier comparison would not surface as naturally.

11.3 Longer-horizon directions

- **Multi-agent extension:** a per-brand or per-segment agent team (mirroring L2M-AID's multi-agent structure, Section 2.2) coordinating response across a simulated multi-PLC network, rather than a single agent observing one flow stream — would let the environment model lateral-movement-style attacks that cross device boundaries, which the current single-stream episode design cannot represent.
- **Online/continual LLM archetype refinement:** currently the 25 archetypes and their LLM risk scores are computed once, offline, from the training split. An online variant that periodically re-clusters and re-scores archetypes as new capture data arrives would test whether the shaping mechanism's benefit persists (or degrades/improves) under distribution shift — relevant to a real ICS deployment where new device firmware or new attack variants appear over time.
- **From simulator to shadow deployment:** the natural next validation step beyond this thesis's simulator-only evaluation is a shadow-mode deployment (policy runs alongside, but does not act on, a real ICS network tap) to test whether the reward function's asymmetric costs and the archetype-based shaping generalize to live traffic characteristics not present in any ICS-NAD capture file.

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